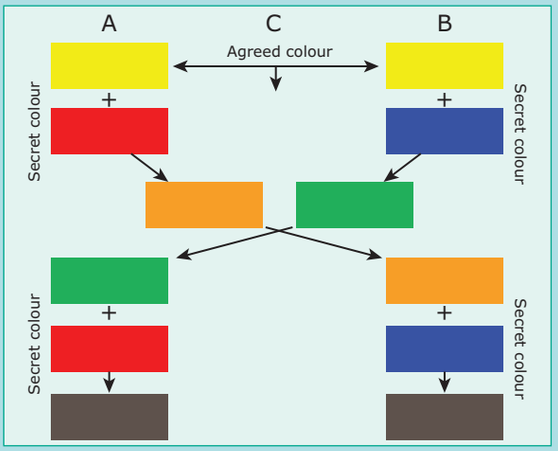
DO YOU KNOW?

In 1976, Whitfield Diffie and Martin e. Hellman, devised an algorithm called public key encryption. The algorithm can be understood using color game. This how could "A" and "B" get a secret key without letting "C" finding it out. The trick is based on 2 facts

- It is easy to mix 2 colors together to get 3rd color
- Given a mixed color it's hard to reverse it in order to find the exact original colors



1. First A and B agree publicly on a starting color (yellow)
2. Now A select a random colour (red) mix it with yellow and send new color (yellow+red=orange) to B.
3. Similarly B selects a random colour (blue) mix it with yellow and send new colour (yellow+blue=green) to A.
4. Hacker "C" may have two new colours (orange) and (green) but not the A's (red) or B's (blue) private colours.
5. After interchanging colors, A adds his own private (red) to B's mixture (green) and arrive at a third secret colour(black).
6. Also B adds his own private (blue) to A's mixture (orange) and arrive at a same third secret color (black).
7. C is unable to have the exact color (black), since C needs one of the private color to do so.

Digital certificates

A digital certificate (also known as public key certificate) is an electronic document used to prove the ownership of a public key. This certificate includes the information about the sender's identity, digital signature and a public key.

A digital certificate function is similar to the function of identification cards such as passports and driving licenses. Digital certificates are issued by recognized Certification Authorities (CA). When someone requests a digital certificate, the authority verifies the identity of the requester, and if the requester fulfills all requirements, the authority issues it. When the sender uses a certificate to sign a document digitally, receiver can trust the digital signature because he trusts that CA has done their part verifying the sender's identity.



Figure 17.2 Digital Certificates

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